Markov Chain Generation with Interval Algorithm

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Contents

Chapter 1

Introduction

For testing purpose, in many fields of science and engineering, we use a computer to simulate natural phenomena rather than experiment with the real system. Applications of random number generation can be found in simulation of physical processes like sampling applied statistics, atomic collisions. Random number generation also plays an important role in cryptography where the requirement of randomness is more stringent

So far, the problem of generating an independent and identically distributed random variables are studied. In this paper, we shall be dealt with a more general problem of generating Makor source of the first order and we will propose two algorithms for generating a Markov sequence. While the first algorithm provides exact Markov source generating, the second algorithm provides approximate a Markov sequence generating. Furthermore, in the first case, evaluation of the number of coin tosses required for generating a Markov sequence is obtained asymptotically[1]. In the second case, the distance between the output of a Markov sequence generated by the algorithm and the standard Markov sequence in term of variational distance is also evaluated asymptotically.

In details, the following results are obtained:

- Theorem
- ...

[1]How to express our results in more details?

Chapter 2

Historical Review

Random number generation is the transformation of a random variable X into another random variable Y with a specified probability distribution. The origin of random number generation problems seems to be first introduced by von Neumann [5]. He considered the simulation problem of an *unbiased* coin using a biased coin with an *unknown* distribution in which the symmetry of the unknown distribution plays a key role.

Roche [8] and Abrahams [9] have formulated the general problem of generating an *arbitrary* targe distribution using a general biased M-coin (cf. 3.0.10) with a *known* distribution. Roche has shown that the minimum expected number of coin tosses required to generate the target distribution, which is the performance of the random number generating algorithm, is generally expressed in terms of the ratio of the entropy of the target distribution to that of the coin distribution. Roche has also shown the *existence* of a good algorithm for random number generating in which required number of coin tosses satisfies that condition.

Bases on the idea that is introduced by Knuth and Yao [10], Han and Hoshi [1] have developed a *deterministic* algorithm for generating random number of general kind called *interval algorithm* . The key idea of the *interval algorithm* is a successive refinement of interval partitions.

Interval algorithm is not only simple but also very flexible. This algorithm, in principle, can be directly applied to the very general situations such that the processes of M-coin tosses and random numbers to be generated could be non-i.i.d.. Han and Hoshi have also succesfully applied the *interval algorithm* for homophomic coding. As can be seen later, *interval algorithm* is also applicable in generating a Markov sequence as well as simulating channels.

Using the *interval algorithm*, T. Uyematsu and F. Kanaya [2] have been constructed two algorithms for simulating a descrete memoryless channel. The first algorithm provides exact channel simulation and the second algorithm provides approximate channel simulation. T. Uyematsu and F. Kanaya [**?**, key3]ave shown that in the first algorithm, the number of fair random bits per input sample approaches the conditional resolvability of channel with probability one, and in the second algorithm, the approximation error measured by the variational distance vanishes exponentially as the blocklength tends to infinity, when *the number of fair random bits per input sample is the above the conditional resolvability*.

In this paper, we will be dealt with the problem of generating the first order of Markove sequence using the *interval algorithm*. We propose two algorithms for generating the first order of Markov sequence. The first algorithm generates *exact* Markov sequence and the second algorithm generates the

Markov sequence approximately. In order to obtain the evaluation for performances of two algorithms, we use the *Markove type*, which is introduced by Davisson *et al* [3] and in the proofs for the theorems we will propose, we use the techniques similar to the techniques used in T. Uyematsu and F. Kanaya [2].

Chapter 3

Notations and Basic Definitions

Let $\mathcal{X} \stackrel{\text{def}}{=} \{1, 2, \cdots, M\}$ and let X be a random variable taking values in $\mathcal{X}$ with probability distribution $P_X(X) \equiv P(X) = \Pr(X = x)$.

DEFINITION 3.0.1 (Entropy). *The entropy of X is defined by*

$$H(X) \equiv H(P_X) \stackrel{\text{def}}{=} -\sum_{x \in \mathcal{X}} P_X(x) \log P_X(x). \tag{3.1}$$

From now on, all logarithms and exponentials are considered to the base of two.

DEFINITION 3.0.2 (Conditional Entropy). *Let Y be a random variable taking values in a finite set $\mathcal{Y}$. The conditional entropy $H(Y|X)$ is defined as*

$$H(Y|X) \stackrel{\text{def}}{=} \sum_{x \in \mathcal{X}} P_X(x) \left[-\sum_{y \in \mathcal{Y}} P(y|x) \log P(y|x) \right] \tag{3.2}$$

DEFINITION 3.0.3 (Divergence). *For probability distributions P, Q taking values in $\mathcal{X}$, the information divergence $D(P||Q)$ is defined by*

$$D(P||Q) \stackrel{\text{def}}{=} \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}. \tag{3.3}$$

Divergence $D(P||Q)$ is always non-negative and $D(P||Q) = 0$ if and only if $P(x) = Q(x)$ for all $x \in \mathcal{X}$.

DEFINITION 3.0.4 (Conditional Divergence). *Let P, Q are probability distributions taking values in $\mathcal{X}$, R is a probability distribution taking values in $\mathcal{Y}$.The conditional divergence $D(P||Q|R)$ is define as*

$$D(P||Q|R) \stackrel{\text{def}}{=} \sum_{y \in \mathcal{Y}} R(y) \sum_{x \in \mathcal{X}} P(x|y) \log \frac{P(x|y)}{Q(x|y)}. \tag{3.4}$$

DEFINITION 3.0.5 (Variational distance). *The variational distance between two distributions P and Q on $\mathcal{X}$ is*

$$d(P, Q) \stackrel{\text{def}}{=} \sum_{x \in \mathcal{X}} |P(x) - Q(x)|. \tag{3.5}$$

DEFINITION 3.0.6 (Markov sequence [1]). *A finite Markov sequence with constant transition probability is a random variable taking values in $\mathcal{X} = \{1, 2, \cdots M\}$ for which*

$$\Pr\{X_i = j_i | X_1 X_2 \cdots X_{i-1} = j_1 j_2 \cdots j_{i-1}\} = \Pr\{X_i = j_i | X_{i-1} = j_{i-1}\}$$

for all possible values of indices.

In this paper, we shall be concerned only with such Markov sequences.

DEFINITION 3.0.7 (Transition matrix). *The transition matrix of a Markov sequence taking values in $\mathcal{X} = \{1, 2, \cdots M\}$ is an $M \times M$ matrix $P = [p_{kj}]_{k,j=1}^{M}$ where $p_{kj} = \Pr\{X_i = k | X_{i-1} = j\}$.*

DEFINITION 3.0.8 (Stationary probability distribution). *The stationary probability distribution $q = (q_1, \cdots, q_M)$ of a Markov chain with transition matrix $P = [p_{kj}]_{k,j=1}^{M}$ is a probability distribution q such that:*

$$qP = q.$$

DEFINITION 3.0.9 (Joint distribution and joint type). *In the sequel we let $\mathcal{P}(\mathcal{X} \times \mathcal{X})$ denotes the set of all joint distribution on $\mathcal{X} \times \mathcal{X}$, and let $\mathcal{P}_n(\mathcal{X} \times \mathcal{X})$ denote the set of all joint types for $\mathcal{X}^n \times \mathcal{X}^n$. It is obviously that $\mathcal{P}_n(\mathcal{X} \times \mathcal{X}) \subset \mathcal{P}(\mathcal{X} \times \mathcal{X})$. Furthermore, we use the convention that, given a conditional distribution $Q = [q_{kj}]_{k,j=1}^{M}$ and a marginal distribution $q = (q_j)_{j=1}^{M}$, $q \circ Q$ indicates the joint distribution $q \circ Q = [q_j q_{kj}]_{k,j=1}^{M} \subset \mathcal{P}(\mathcal{X} \times \mathcal{X})$.*

DEFINITION 3.0.10 (N-coin). *In general,an N-coin is a random variable Z taking values in $\mathcal{Z} = \{1, 2, \cdots, N\}$ with probability $\mathbf{z} = \{z_1, \cdots, z_N\}$(i.e. $\Pr\{Z = i\} = z_i, \forall i \in \mathcal{Z}$. Without loss of generality, we assume that: $z_i > 0 \quad$ (for all $i \in \mathcal{Z}$). In the case of $z_1 = z_2 = \cdots = z_N = 1/N$, the coin is said to be unbiased, otherwise it is biased.*

[1]Ask Prof. Han

Chapter 4

Makov type of sequences

Let $\mathbf{x} = x_1 x_2 \cdots x_n$ be a sequence where $x_i \in \mathcal{X} = \{1, 2, \cdots M\}, \quad i = \overline{1, n}$.
Let n_{kj} be the number of transitions from j to k in $\mathbf{x}$,i.e. the number of ordered pairs ij in $\mathbf{x}$ with the *cyclic* convention that x_n precedes x_1.

DEFINITION 4.0.11 (Markov type of a sequence). *The matrix $[n_{kj}/n_j]_{k,j=1}^{M}$ is called the Markov type of the sequence $\mathbf{x}$, where*

$$n_j = \sum_{k=1}^{M} n_{kj}. \tag{4.1}$$

DEFINITION 4.0.12 (Entropy of a Markov type).

$$H(Q|q) \stackrel{\text{def}}{=} \sum_{j=1}^{M} q_j H(Q^{(j)}) \tag{4.2}$$

$$= -\sum_{j=1}^{M} \frac{n_j}{n} \sum_{k=1}^{M} \frac{n_{kj}}{n_j} \log \frac{n_{kj}}{n_j} = -\frac{1}{n} \sum_{j=1}^{M} \sum_{k=1}^{M} n_{kj} \log \frac{n_{kj}}{n_j}, \tag{4.3}$$

where

$$q_j = n_j/n, \quad j = \overline{1, M}, \tag{4.4}$$

$$Q^{(j)} = n_{kj}/n_j, \quad k, j = \overline{1, M}. \tag{4.5}$$

DEFINITION 4.0.13 (Divergence of a Markov type).

$$D(Q||P|q) \stackrel{\text{def}}{=} \sum_{j=1}^{M} q_j D(Q^{(j)}||P^{(j)}) \tag{4.6}$$

$$= \sum_{j=1}^{M} q_j \sum_{k=1}^{M} q_{kj} \log \frac{q_{kj}}{p_{kj}} \tag{4.7}$$

$$= \sum_{j=1}^{M} \frac{n_j}{n} \sum_{k=1}^{M} \frac{n_{kj}}{n_j} \log \frac{n_{kj}/n_j}{p_{kj}} \tag{4.8}$$

$$= \sum_{j=1}^{M} \sum_{k=1}^{M} \frac{n_{kj}}{n} \log \frac{n_{kj}/n_j}{p_{kj}} \tag{4.9}$$

where $q^{(j)} = (q_{kj})_{k=1}^{M}$ and $P^{(j)}$ is the transition matrix; $P^{(j)} = (p_{kj})_{k=1}^{M}$.

The following results are proved by Davisson *et al*[3]:

LEMMA 4.0.14. *Let* $\mathbf{x} = x_1 x_2 \cdots x_n$ *be a Markov sequence taking values in* $\mathcal{X} = \{1, 2, \cdots, M\}$ *with transition matrix* $P = [p_{kj}]_{k,j=1}^{M}$ *and stationary probability distribution* $q = (q_j)_{j=1}^{M}$. *We assume that:* $p_{kj} > 0 \quad \forall j, j = \overline{1, M}$ *and* $q_j > 0 \quad \forall j = \overline{1, M}$.

Let $\alpha = \min_j q_j$ *and* $\beta = \min_{k,j} p_{kj}$.

Let $T = [n_{kj}/n_j]_{k,j=1}^{M}$ *be a Markov type. We set* $Q = [n_{kj}/n]_{k,j=1}^{M}$. *Let* $\mathbf{x} = x_1 x_2 \cdots x_n \in \mathcal{X}^n(T)$. *We have*

- $\Pr(\mathbf{x}) = \frac{p_{x_1}}{p_{x_n x_1}} \exp\left[-nD(Q||P|q) - nH(Q|q)\right]$.
- $n^{-M}(n+1)^{-M^2} \exp(nH(Q|q)) \leq |\mathcal{X}^n(T)| \leq M \exp(nH(Q|q))$.
- $n^{-M}(n+1)^{-M^2} \alpha \exp(-nD(Q||P|q)) \leq \Pr\{\mathcal{X}^n(T)\} \leq (M/\beta) \exp(-nD(Q||P|q))$

where $|\cdot|$ denotes the cardinity of the set and $\mathcal{X}^n(T)$ denotes the set of all sequences with the same type T.

Chapter 5

Statements of the problems

In this paper, we will be dealt with the following problems:

5.1 Problem 1: Generating exactly a Markov sequence with an N-coin using the interval algorithm

Given a Markov sequence $\mathbf{x} = x_1 x_2 \cdots x_n$ where $\mathbf{x} \in \mathcal{X}^n = \{1, 2, \cdots M\}^n$. Let the transition matrix and the stationary distribution of the Markov sequence be as follows, respectively

$$p_{kj} = \Pr\{X_2 = k | X_1 = j\}, \quad (k = \overline{1, M}, j = \overline{1, M}), \tag{5.1}$$

$$q_j = \Pr\{X_1 = j\} \quad (j = \overline{1, M}). \tag{5.2}$$

We set $P = [p_{kj}]_{k,j=1}^M$ and $q = (q_j)_{j=1}^M$.

Let $T = [n_{kj}/n_j]_{k,j=1}^M$ be a Markov type and let $\mathbf{x} \in \mathcal{X}^n(T)$ where $\mathcal{X}^n(T)$ denotes the set of all sequences with sampe type T. For the sake of simplicity, we also set $Q = [n_{kj}/n]_{k,j=1}^M$.

Propose an *interval algorithm* for generating the Markov sequence X^n using a general unbiased N-coin Z with probability $\mathbf{z} = \{z_1, z_2, \cdots, z_N\}$.

Furthermore, in the case of $N = 2$, $\quad z_1 = z_2 = 1/2$, evaluate the number of coin tosses needed to generate the Markov sequence $\mathbf{x}$.

5.2 Problem 2: Generating approximately a Markov sequence with an N-coin using the interval algorithm

Given a Markov sequence $\mathbf{x} = x_1 x_2 \cdots x_n$ where $\mathbf{x} \in \mathcal{X}^n = \{1, 2, \cdots M\}^n$. Let the transition matrix and the stationary distribution of the Markov sequence be as follows, respectively

$$p_{kj} = \Pr\{X_2 = k | X_1 = j\}, \quad (k = \overline{1, M}, j = \overline{1, M}), \tag{5.3}$$

$$q_j = \Pr\{X_1 = j\} \quad (j = \overline{1, M}). \tag{5.4}$$

We define $P = [p_{kj}]_{k,j=1}^M$ and $q = (q_j)_{j=1}^M$ respectively.

Let $T = [n_{kj}/n_j]_{k,j=1}^M$ be a Markov type and let $\mathbf{x} \in \mathcal{X}^n(T)$ where $\mathcal{X}^n(T)$ denotes the set of all sequences with sampe type T. For the sake of simplicity, we also set $Q = [n_{kj}/n]_{k,j=1}^M$.

Propose an *interval algorithm* for generating the Markov sequence X^n using a general unbiased N-coin Z with probability $\mathbf{z} = \{z_1, z_2, \cdots z_N\}$ when the N-coin is tossed $K = nR$ times.

Furthermore, in the case of $N = 2, \quad z_1 = z_2 = 1/2$, evaluate the approximation error of the algorithm measured by the variational distance (cf. 3.0.5, pp. 5) .

Chapter 6

N-coin tosses for generating a Markov sequence

6.1 Interval algorithm for generating a Markov sequence with a N-coin

Here, we will use the notations X^n, Z, P, q and Q as defined in section 5.1 (pp. 9).

Iterative Algorithm for Generating a Markov sequence:

1) Set $m = 1, \quad s = t = \lambda$ (null string), $\quad \alpha_s = \gamma_t = 0, \quad \beta_s = \delta_t = 1, \quad I(s) = [\alpha_s, \beta_s)$, and $J(t) = [\gamma_t, \delta_t)$.

2a) Partition the unit inverval $[0, 1)$ into M disjoint subintervals $J(1), \cdots, J(M)$ such that

$$J(l) = [Q_{l-1}, Q_l)\,, \quad (l = \overline{1, M}) \tag{6.1}$$

where

$$Q_l = \sum_{k=1}^{l} q_k \quad (l = \overline{1, M}; \quad Q_0 = 0). \tag{6.2}$$

2b) Set

$$P_l = \sum_{k=1}^{l} z_k \quad (l = \overline{1, N}; \quad P_0 = 0). \tag{6.3}$$

3) Toss the N-coin Z with probability $\mathbf{z} = \{z_1, \cdots, z_N\}$ to have a value $a \in \{1, \cdots, N\}$, and generate the subinterval of $I(s)$

$$I(sa) = [\alpha_{sa}, \beta_{sa}) \tag{6.4}$$

where

$$\alpha_{sa} = \alpha_s + (\beta_s - \alpha_s)P_{a-1}, \tag{6.5}$$
$$\beta_{sa} = \alpha_s + (\beta_s - \alpha_s)P_a. \tag{6.6}$$

4a) If $I(sa)$ is entirely contained in some $J(ti)$, $i = \overline{1, M}$, then output i as the value of the m-th output X_m and set $t = ti$. Otherwise, go to 5).

4b) If $m = n$ then stop the algorithm. Otherwise, partition the interval $J(t) \equiv [\gamma_t, \delta_t)$ into M disjoint subintervals $J(t1), \cdots, J(tM)$ such that

$$J(tj) = [\gamma_{tj}, \delta_{tj}) \quad (j = \overline{1, M}); \tag{6.7}$$

where

$$\gamma_{tj} = \gamma_t + (\delta_t - \gamma_t)Q_{j-1}^{(m)}; \tag{6.8}$$
$$\delta_{tj} = \gamma_t + (\delta_t - \gamma_t)Q_j^{(m)}, \tag{6.9}$$

with

$$Q_h^{(m)} = \sum_{j=1}^{h} p_{ji} \quad (h = \overline{1, M}; \quad Q_0^{(m)} = 0), \tag{6.10}$$

set $m := m + 1$ and go to 4a).

5) Set $s = sa$ and go to 3).

In the sequel, we consider the case in which the coin is unbiased with $N = 2$ and $z_1 = z_2 = 1/2$. (This case is called the **resolvability** problem).

Now let T_n be the random variable indicating the number of coin tosses needed to generate a Markov sequence $\mathbf{x} = x_1 \cdots x_n$ in $\mathcal{X}^n$. We have the following theorem:

6.2 Theorem 6.2.1

THEOREM 6.2.1. *The coin tosses number T_n needed to generate a Markov sequence $\mathbf{x} = x_1 \cdots x_n$ in $\mathcal{X}^n$ satisfies:*

$$\liminf_{n\to\infty} \left[-\frac{1}{n} \log \Pr\{T_n \geq nR\} \right] \geq E_{\mathrm{r}}(R, P, q) \tag{6.11}$$

with

$$E_{\mathrm{r}}(R, P, q) \stackrel{\text{def}}{=} \min_{\substack{Q \in V(\mathcal{X}): \\ q \circ Q \in P(\mathcal{X} \times \mathcal{X}), qQ = q}} \left(D(Q||P|q) + |R - H(Q|q) - D(Q||P|q)|^+ \right) \tag{6.12}$$

where $|x|^+ \stackrel{\text{def}}{=} \max\{0, x\}$.

Furthermore $E_{\mathrm{r}}(R, P, q) > 0$ *if and only if* $R > H(P|q)$ *where* P *is the transition matrix and* q *is the marginal distribution of the Markov sequence* $\mathbf{x}$*, respectively .*

On the other hand,

$$\lim_{n\to\infty} \left[-\frac{1}{n} \log \Pr\{T_n \leq nR\} \right] = F(R, P, q) \tag{6.13}$$

where

$$F(R, P, q) \stackrel{\text{def}}{=} \min_{\substack{Q \in V(\mathcal{X}): \\ q \circ Q \in P(\mathcal{X} \times \mathcal{X}), \\ D(Q||P|q) + H(Q|q) \leq R}} D(Q||P|q) \tag{6.14}$$

$F(R, P, q)$ *may be* $+\infty$*. Furthermore,* $F(R, P, q) > 0$ *if and only if* $R < H(P|q)$ *.*

Remark: The meaning of this theorem can be ”translated” roughly as follows

$$\liminf_{n\to\infty} \left[-\frac{1}{n} \log \Pr\{T_n \geq nR\} \right] \geq E_{\mathrm{r}}(R, P, q) \tag{6.15}$$

implies that there exists infinite n such that

$$-\frac{1}{n} \log \Pr\{T_n \geq\} \geq E_{\mathrm{r}}(R, P, q) \tag{6.16}$$

and hence

$$\Pr\{T_n \geq\} \leq \exp\left[-nE_{\mathrm{r}}(R, P, q)\right] \tag{6.17}$$

therefore, if $E_{\mathrm{r}}(R, P, q) \geq 0$, the probability that the number of coin tosses needed to generate $\mathbf{x}$, exceeds nR vanishes exponentially as n n to inifinity.

Chapter 7

Generating a Markov sequence with a fixed number of coin tosses

7.1 Interval Algorithm for generating a Markov sequence with a fixed number of coin tosses

Now we consider another problem when the number of coin tosses is fixed. In this case, we cannot generate the Markov sequence exactly but approximately. First, we modify the interval algorithm for generating a Markov equence such that the algorithm outputs a *dummy sequence* $x_1x_2\cdots x_n = 11\cdots 1$, whenever the algorithm does not stop within $K = nR$ coin tosses.

Here, we use the notations X^n, Z, P, q and Q as defined in section 5.2 (pp. 9).

Iterative Algorithm for Generating a Markov sequence with $K = nR$ coin tosses:

1) Set $l = 0$, $s = t = \lambda$(null string), $\alpha_s = \gamma_t = 0$, $\beta_s = \delta_t = 1$, $I(s) = [\alpha_s, \beta_s)$, and $J(t) = [\alpha_s, \beta_s)$.

2a) Partition the unit interval $[0, 1)$ into M disjoint subintervals $J(1), \cdots, J(M)$ such that

$$J(l) = [Q_{l-1}, Q_l)\,, \quad (l = \overline{1, M}); \tag{7.1}$$

where

$$Q_l = \sum_{k=1}^{l} q_k, \quad (l = \overline{1, N}; \quad Q_0 = 0). \tag{7.2}$$

2b) Set

$$P_l = \sum_{k=1}^{l} z_k \quad (l = \overline{1, N}; \quad P_0 = 0). \tag{7.3}$$

3) If $l = K$, generate the output $11\cdots 1$ as the output of sequence $\mathbf{x}$ and stop the algorithm. Otherwise, toss the N-coin Z which takes values on $\{1, \cdots, N\}$ with probability $\mathbf{z} = \{z_1, \cdots, z_N\}$ to have a random number a and generate the subinterval of $I(s)$:

$$I(sa) = [\alpha_{sa}, \beta_{sa})$$

where

$$\alpha_{sa} = \alpha_s + (\beta_s - \alpha_s)P_{a-1}; \tag{7.4}$$
$$\beta_{sa} = \alpha_s + (\beta_s - \alpha_s)P_a. \tag{7.5}$$

Set $l := l + 1$.

4a) If $I(sa)$ is entirely contained in some $J(ti), (i = \overline{1, M}$), set $t = ti$. Otherwise go to 5).

4b) If $m = n$, generate output t as the output of sequence $\mathbf{x}$ and stop the algorithm. Otherwise, partition the interval

$$J(t) = [\gamma_t, \delta_t) \tag{7.6}$$

into M disjoint subintervals $J(t1), \cdots J(tM)$ such that

$$J(tj) = [\gamma_{tj}, \delta_{tj}), \quad (j = \overline{1, M}); \tag{7.7}$$

where

$$\gamma_{tj} = \gamma_t + (\delta_t - \gamma_t)Q_{j-1}^{(m)}; \tag{7.8}$$
$$\delta_{tj} = \gamma_t + (\delta_t - \gamma_t)Q_j^{(m)}, \tag{7.9}$$

with

$$Q_h^{(m)} = \sum_{j=1}^{h} p_{ji} \quad (h = \overline{1, M}; \quad Q_0^{(m)} = 0); \tag{7.10}$$

set $m = m + 1$ and go to 4a).

5) Set $s = sa$ and go to 3).

In what follows, we consider the case of $N = 2$ with $z_1 = z_2 = 1/2$. (This case is called the **resolvability problem**).

7.2 Theorem 7.2.1

THEOREM 7.2.1. *Let $\tilde{P}^n(\mathbf{x})$ denotes the probability of an output sequence $\mathbf{x} \in \mathcal{X}^n$ generated by the Interval Algorithm with nR coin tosses. Then we have:*

$$\liminf_{n\to\infty} \left[-\frac{1}{n} \log d(P_X^n, \tilde{P}_X^n) \right] \geq E_{\mathrm{r}}(R, P, p) \tag{7.11}$$

where $d(P, Q)$ denotes the variational distance between two probability distributions P and Q(cf. 3.0.5, pp. 5) and the function $E_{\mathrm{r}}(R, P, p)$ is given by 6.12, (pp. 12).

Converse version of *themrem 7.2.1*:

7.3 Theorem 7.3.1

[Draft]

THEOREM 7.3.1 (Draft). *For any $\tilde{X}^n \in \mathcal{X}^n$ with its resolution $R(\tilde{X}^n) = nR$, we have*

$$\limsup_{n\to\infty} \left[-\frac{1}{n} \log d(P_X^n, \tilde{P}_X^n) \right] \leq E_{\mathrm{sp}}(R, P) \tag{7.12}$$

where

$$E_{\mathrm{sp}}(R, P, q) \stackrel{\mathrm{def}}{=} \min_{\substack{Q\in V(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)\geq R}} D(Q||P|q). \tag{7.13}$$

$E_{\mathrm{sp}}(R, P, q)$ *may be* $+\infty$*, furthermore,*

$$E_{\mathrm{sp}}(R, P, q) \geq E_{\mathrm{r}}(R, P, q) \tag{7.14}$$

and equality holds for $R \leq R_o$ *where*

$$R_o \stackrel{\mathrm{def}}{=} D(Q_o||P|q) + \log M \tag{7.15}$$

and $P_x(b) \stackrel{\mathrm{def}}{=} 1/M$ *for every* $b \in \mathcal{X}$*(* $P_x(b)$ *is the type of b).*

7.4 Theorem 7.4.1

[Draft]

THEOREM 7.4.1. *Let $\tilde{P}_X^n$ denotes the probability of an output sequence $X^n \in \mathcal{X}^n$ generated by the interval algorithm with nR coin tosses. Then*

$$\lim_{n\to\infty} \left[-\frac{1}{n} \log \left(2 - d(P_X^n, \tilde{P}_X^n) \right) \right] = F(R, P, q). \tag{7.16}$$

where the function $F(R, P, q)$ is given by6.14

7.5 Theorem 7.5.1

[Draft]

THEOREM 7.5.1 (Draft). *Let $\bar{P}_X^n$ denotes the probability which minimizes the variational distance $d(P_X^n, \bar{P}_X^n)$ under the condition $R(\bar{X}^n) = nR$. Then*

$$\lim_{n\to\infty} \left[-\frac{1}{n} \log \left(2 - d(P_X^n, \bar{P}_X^n) \right) \right] = G(R, P, q) \tag{7.17}$$

where

$$G(R,P,q) \stackrel{\text{def}}{=} \min_{\substack{Q\in V(\mathcal{X}):\\ H(Q|q)\leq R}} D(Q||P|q) \tag{7.18}$$

Obviously

$$G(R,P,q) < F(R,P,q), \textit{for } R < H(P|q) \tag{7.19}$$

Therefore, the interval algorithm with nR coin tosses cannot achieve the optimum exponent, whenever $R < H(P|q)$.

Chapter 8

Proofs of Theorems

8.1 Proof of themrem 6.2.1

We correspond each sequence $\mathbf{x} = x_1 x_2 \cdots x_n \in \mathcal{X}^n$ to several distinct subintervals $J(\mathbf{x})$ of $[0, 1)$ with width $P^n(\mathbf{x})$. On the other hand, partition the unit interval $[0, 1)$ into $\exp(nR)$ subintervals

$$I_i \stackrel{\text{def}}{=} [(i-1)\exp(-nR), i\exp(-nR)) , \quad i = \overline{1, \exp(-nR)}.$$

Then, the interval I_i, corresponds to each outcome of nR coin tosses. If the subinterval I_i is completely included in some $J(\mathbf{x})$, then the sequence of coin tosses corresponding to I_i can terminate the algorithm. By using this observation, we have

$$\Pr\{T_n \geq nR\} \leq \sum_{\substack{\mathbf{x}\in\mathcal{X}^n: \\ P^n(\mathbf{x})\geq\exp(-nR)}} 2\exp(-nR) + \sum_{\substack{\mathbf{x}\in\mathcal{X}^n: \\ P^n(\mathbf{x})<\exp(-nR)}} P^n(\mathbf{x}). \tag{8.1}$$

For the sake of simplicity, put $\gamma_n = \frac{p_{x_1}}{p_{x_n x_1}}$ and $\omega_n = \frac{1}{n}\log\left[\frac{p_{x_1}}{p_{x_n x_1}}\right]$.
We have:

$$P^n(\mathbf{x}) \geq \exp(-nR) \Leftrightarrow \tag{8.2}$$
$$\gamma_n \exp\bigl[-nD(Q||P|q) - nH(Q|q)\bigr] \geq \exp(-nR) \Leftrightarrow \tag{8.3}$$
$$-n(D(Q||P|q) + H(Q|q)) \geq -nR - \log\gamma_n \Leftrightarrow \tag{8.4}$$
$$D(Q||P|q) + H(Q|q) \leq R + \omega_n. \tag{8.5}$$

Remark: It can be seen that

$$\frac{p_{x_1}}{\max_{kj} p_{kj}} \leq \gamma_n \leq \frac{p_{x_1}}{\min_{kj} p_{kj}}$$

i.e. the sequence $\{\gamma_n\}$ is bounded and therefore

$$\lim_{n\to\infty} \omega_n = 0.$$

Similarly, we have:

$$
\begin{aligned}
P^n(\mathbf{x}) < \exp(-nR) \Leftrightarrow && (8.6)\\
\gamma_n \exp\big[-nD(Q||Q|q) - nH(Q|p)\big] < \exp(-nR) \Leftrightarrow && (8.7)\\
-n(D(Q||P|q) + H(Q|q)) < -nR - \log\gamma_n \Leftrightarrow && (8.8)\\
D(Q||P|q) + H(Q|q) > R + \omega_n. && (8.9)
\end{aligned}
$$

Therefore:

$$
\Pr\{T_n \geq nR\} \leq \sum_{\substack{\mathbf{x}\in\mathcal{X}^n, Q\in V_n(\mathcal{X}):T_{\mathbf{x}}=Q,\\ D(Q||P|q)+H(Q|q)\leq R+\frac{1}{n}\log\omega_n}} 2\exp(-nR) + \sum_{\substack{\mathbf{x}\in\mathcal{X}^n, Q\in V_n(\mathcal{X}):T_{\mathbf{x}}=Q,\\ Q\in V_n(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)<R+\omega_n}} P^n(\mathbf{x}) \quad (8.10)
$$

where $T_{\mathbf{x}}$ denotes the type of the sequence $\mathbf{x}$ and $V_n(\mathcal{X})$ is the set of all Markov type.

We recall that

$$
|\{\mathbf{x}\in\mathcal{X}^n : T_{\mathbf{x}} = Q\}| \leq M\exp\left(nH(Q|q)\right). \quad (8.11)
$$

We have:

$$
\begin{aligned}
\Pr\{T_n \geq nR\} \leq & \sum_{\substack{Q\in V_n(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)\leq R+\omega_n}} \Big\{2\exp(-nR)\Big\}\Big\{M\exp\Big[nH\left(Q|q\right)\Big]\Big\}\\
& + \sum_{\substack{Q\in V_n(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)>R+\omega_n}} \gamma_n\exp\Big[-n\big\{D(Q||P|q)+H(Q|q)\big\}\Big] \quad (8.12)
\end{aligned}
$$

$$
\begin{aligned}
= & \sum_{\substack{Q\in V_n(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)\leq R+\omega_n}} 2M\exp\gamma_n\exp\left[-n\left\{R+\omega_n - H(Q|q)\right\}\right]\\
& + \gamma_n \sum_{\substack{Q\in V_n(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)>R+\omega_n}} \exp\Big[-n\left\{D(Q||P|q)+H(Q|q)\right\}\Big] \quad (8.13)
\end{aligned}
$$

$$
\begin{aligned}
\leq \max\Big\{2M, \sup_n\big\{\gamma_n\exp\gamma_n\big\}\Big\} & \\
\Bigg\{ & \sum_{\substack{Q\in V_n(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)\leq R+\omega_n}} \exp\left[-n(R+\omega_n - H(Q|q))\right]\\
& + \sum_{\substack{Q\in V_n(\mathcal{X}):\\ D(Q||P|q)+H(Q|q)>R+\omega_n}} \exp\Big[-n\left(D(Q||P|q)+H(Q|q)\right)\Big]\Bigg\} \quad (8.14)
\end{aligned}
$$

$$\leq \max\left\{2M, \sup_n \left\{\gamma_n \exp \gamma_n\right\}\right\} \sum_{Q \in V_n(\mathcal{X})} \exp\left[-n\left(D(Q||P|q) + |R + \omega_n - H(Q|q) - D(Q||P|q)|^+\right)\right] \tag{8.15}$$

$$\leq \max\{2M, \sup_n\{\gamma_n \exp \gamma_n\}\}(n+1)^{M^2} \exp\Big[-nE_{\mathrm{r}}(R + \omega_n, P, q)\Big] \tag{8.16}$$

Now we can write:

$$\Pr\{T_n \geq nR\} \leq \max\left\{2M, \sup_n \left\{\gamma_n \exp \gamma_n\right\}\right\}$$
$$\sum_{Q \in V_n(\mathcal{X})} \exp\Big[-n\left\{D(Q||P|q) + |R + \omega_n - H(Q|q) - D(Q||P|q)|^+\right\}\Big] \tag{8.17}$$

$$\leq \quad \max\left\{2M, \sup_n \left\{\gamma_n \exp \gamma_n\right\}\right\}(n \quad + \quad 1)^{M^2} \exp\left[-nE_{\mathrm{r}}(R + \omega_n, P, q)\right] \tag{8.18}$$

Hence

$$-\frac{1}{n} \log \Pr\{T_n > nR\} \geq -\frac{\log \max\left\{2M, \sup_n \left\{\gamma_n \exp \gamma_n\right\}\right\}}{n} - M^2 \frac{\log(n+1)}{n} + E_{\mathrm{r}}(R + \omega_n, P, q). \tag{8.19}$$

where $E_{\mathrm{r}}(R, P, q)$ is given in 6.12 pp. 12.
We have

$$\lim_{n \to \infty} \frac{\log(n+1)}{n} = 0, \tag{8.20}$$

and

$$\lim_{n \to \infty} \frac{\log \max\left\{2M, \sup_n \left\{\gamma_n \exp \gamma_n\right\}\right\}}{n} = 0, \tag{8.21}$$

therefore we can conclude that:

$$\liminf_{n \to \infty} \left[-\frac{1}{n} \log \Pr\{T_n \geq nR\}\right] \geq E_{\mathrm{r}}(R, P, q). \tag{8.22}$$

On the other hand, $E_{\mathrm{r}}(R, P, q) = 0$ if and only if $Q \in V(\mathcal{X})$ satisfies $Q = P$(hence $D(Q||P|q) = 0$) and $D(Q||P|q) + H(Q|q) \geq R$ or $H(P|q) \geq R$ because $D(Q||P|q) = 0$ under the condition

$Q = P$.

So we can conclude that $E_{\mathrm{r}}(R, P, q) > 0$ if and only if $R > H(P|q)$.

Now we will show the second part of 6.2.1. Note that when $P^n(\mathbf{x}) < \exp(-nR)$, the length of I_i is greater than $P^n(\mathbf{x})$ and therefore I_i can not be completely included in $J(\mathbf{x})$.

Process in a similar manner to the first part, we have:

$$\begin{aligned}
\Pr\{T_n \leq nR\} \leq & \\
& \sum_{\mathbf{x}: P^n(\mathbf{x}) \geq \exp(-nR)} P^n(\mathbf{x}) \\
&= \sum_{\substack{\mathbf{x}: T_{\mathbf{x}} = Q, Q \in V_n(\mathcal{X}): \\ D(Q||P|q) + H(Q|q) \leq R + \omega_n}} P^n(\mathbf{x}) \\
\leq & \sum_{\substack{Q \in V_n(\mathcal{X}): \\ D(Q||P|q) + H(Q|q) \leq R + \omega_n}} \left\{ M \exp\left[nH(Q|q)\right] \right\} \left(\gamma_n \exp\left[-n(D(Q||P|q) + H(Q|p)) \right] \right) \\
&= \sum_{\substack{Q \in V_n(\mathcal{X}): \\ D(Q||P|q) + H(Q|q) \leq R + \omega_n}} M \gamma_n \exp\left[-nD(Q||P|q) \right] \qquad (8.23)
\end{aligned}$$

and therefore:

$$\Pr\{T_n \leq nR\} \leq M \gamma_n (n+1)^{M^2} \exp\left[-nF(R + \omega_n, P, q) \right]. \qquad (8.24)$$

Take logarithm of both sides, we have

$$-\frac{1}{n} \log \Pr\{T_n \leq nR\} \geq -\frac{\log M\,(\gamma_n)}{n} - M^2 \frac{\log(n+1)}{n} + F(R + \omega_n, P, q) \qquad (8.25)$$

Therefore

$$\liminf_{n \to \infty} \left[-\frac{1}{n} \log \Pr\{T_n \leq nR\} \right] \geq F(R, P, q). \qquad (8.26)$$

Similarly,

$$\Pr\{T_n < nR\} \geq \frac{1}{3} \sum_{\substack{\mathbf{x} \in \mathcal{X}^n: \\ P^n(\mathbf{x}) \geq 2\exp(-nR)}} P^n(\mathbf{x}) \qquad (8.27)$$

$$= \frac{1}{3} \sum_{\substack{\mathbf{x} \in \mathcal{X}^n, Q \in V_n(\mathcal{X}), T_{\mathbf{x}} = Q: \\ D(Q||P|q) + H(Q|q) \leq R - \frac{1}{n} + \omega_n}} \gamma_n \exp\left[-n\left(D(Q||P|q) + H(Q|q) \right) \right] \qquad (8.28)$$

We recall that

$$|\{\mathbf{x} \in \mathcal{X}^n : T_{\mathbf{x}} = Q\}| \geq n^{-M}(n+1)^{-M^2} \exp\big[nH(Q|q)\big]. \tag{8.29}$$

and apply this inequality to 8.28, we have

$$\Pr\{T_n \leq nR\} \geq \frac{1}{3} \sum_{\substack{Q \in V_n(\mathcal{X}): \\ D(Q||P|q)+H(Q|q)\leq \\ R-\frac{1}{n}+\omega_n}} \left\{ n^{-M}(n+1)^{-M^2} \exp\Big[nH(Q|q)\Big]\right\}\left\{\gamma_n \exp\Big[-nD(Q||P|q) - nH(Q|q)\Big]\right\} \tag{8.30}$$

$$\geq \frac{1}{3}\inf_n\{\gamma_n\}\, n^{-M}(n+1)^{-M^2} \sum_{\substack{Q \in V_n(\mathcal{X}): \\ D(Q||P|q)+H(Q|q)\leq R-\frac{1}{n}+\omega_n}} \exp\Big[-nD(Q||P|q)\Big] \tag{8.31}$$

$$\geq \frac{1}{3}\inf_n\{\gamma_n\}\, n^{-M}(n+1)^{-M^2} \exp\left[-n \min_{\substack{Q \in V_n(\mathcal{X}): \\ D(Q||P|q)+H(Q|p)\leq R-1/n+\omega_n}} D(Q||P|q)\right] \tag{8.32}$$

$$= \frac{1}{3}\inf_n\{\gamma_n\}\, n^{-M}(n+1)^{-M^2} \exp\left[-nF(R-\frac{1}{n}+\omega_n, P, q)\right], \tag{8.33}$$

therefore

$$-\frac{1}{n}\log \Pr\{T_n \leq nR\} \leq \frac{\log 3 \inf_n\{\gamma_n\}}{n} - M\frac{\log n}{n} - M^2\frac{\log(n+1)}{n} + F(R-\frac{1}{n}+\omega_n, P, q). \tag{8.34}$$

Noting that $F(R,P,q)$ is a continuous function of R therefore $F(R-\frac{1}{n}+\omega_n, P, q)$ tends to $F(R,P)$ when n tends to ∞ and by putting n to ∞ we have

$$\limsup_{n\to\infty}\left[-\frac{1}{n}\log \Pr\{T_n \leq nR\}\right] \leq F(R,P,q). \tag{8.35}$$

Combines (8.26) and (8.35) we have the second part of Theorem 6.2.1 proved.

$$\lim_{n\to\infty}\left[-\frac{1}{n}\log \Pr\{T_n \leq nR\}\right] \leq F(R,P). \tag{8.36}$$

Furthermore, $F(R,P,q) = 0$ if and only if $Q = P$ (thus $D(Q||P|q) = 0$) and $D(Q||P|q|) + H(Q|q) \leq R$ or $H(Q|q) \leq R$. So we can conlude that $F(R,P) > 0$ if and only if $R < H(P|q)$.

8.2 Proof of theorem 7.2.1

Let $\tilde{P}^n(\mathbf{x})$ be the probability of obtaining the output $\mathbf{x} \in \mathcal{X}^n$ by the interval algorithm with nR con tosses.

Remark 1: The output of the algorithm is $11\cdots 1$ if the algorithm stops with $k < nR$ coin tosses

Remark 2: If the algorithm stops with $k < nR$ coin tosses, then the interval coresponding to $\tilde{P}^n(\mathbf{x})$ is a subinterval of the interval coresponding to $P^n(\mathbf{x})$. Hence in this case, $P^n(\mathbf{x}) \geq \tilde{P}^n(\mathbf{x})$.

Remark 3: Furthermore, if the algorithm stops with $k < nR$ coin tosses and $P^n(\mathbf{x}) \geq \exp(-nR)$ then $P^n(\mathbf{x}) < 2\exp(-nR)$ must be held otherwise $P^n(\mathbf{x}) \geq 2\exp(-nR)$ holds and the algorithm has stopped in previous step.

We have

$$0 \leq P^n(\mathbf{x}) - \tilde{P}^n(\mathbf{x}) \tag{8.37}$$

$$\leq \begin{cases} 2\exp(-nR) & \text{,if } P^n(\mathbf{x}) \geq \exp(-nR) \\ P^n(\mathbf{x}) & \text{,Otherwise} \end{cases} \tag{8.38}$$

Using above inequalities and note that

$$\sum_{\mathbf{x}\in\mathcal{X}^n} \tilde{P}^n(\mathbf{x}) = \sum_{\mathbf{x}\in\mathcal{X}^n} P^n(\mathbf{x}) = 1 \tag{8.39}$$

we have,

$$\sum_{\mathbf{x}\in\mathcal{X}^n} \left|\tilde{P}^n(\mathbf{x}) - P^n(\mathbf{x})\right| = \sum_{\substack{\mathbf{x}\in\mathcal{X}^n:\\ x\neq 11\cdots 1}} \left|\tilde{P}^n(\mathbf{x}) - P^n(\mathbf{x})\right| + \left|\sum_{\substack{\mathbf{x}\in\mathcal{X}^n:\\ x\neq 11\cdots 1}} \left(\tilde{P}^n(\mathbf{x}) - P^n(\mathbf{x})\right)\right| \tag{8.40}$$

$$\leq 2 \sum_{\substack{\mathbf{x}\in\mathcal{X}^n:\\ x\neq 11\cdots 1}} \left|\tilde{P}^n(\mathbf{X}) - P^n(\mathbf{x})\right| \tag{8.41}$$

$$\leq 4 \sum_{\substack{\mathbf{x}\in\mathcal{X}^n:\\ P^n(\mathbf{x})\geq\exp(-nR)}} \exp(-nR) + 2 \sum_{\substack{\mathbf{x}\in\mathcal{X}^n:\\ P^n(\mathbf{x})<\exp(-nR)}} P^n(\mathbf{x}) \tag{8.42}$$

Therefore, using the techniques similar to 8.3-8.7, pp. 19, 8.7-8.9,pp. 20 in the previous theorem, we have

$$-\frac{1}{n}\log d(P^n, \tilde{P}^n) \geq -\frac{\log\left(2\max\left\{\gamma_n \exp\gamma_n\right\}\right)}{n} - M^2\frac{\log(n+1)}{n} + E_{\mathrm{r}}(R,P,q) \tag{8.43}$$

Thus

$$\limsup_{n\to\infty}\left[-\frac{1}{n}\log d(P^n, \tilde{P}^n)\right] \geq E_{\mathrm{r}}(R,P) \tag{8.44}$$

This completes the proof of *Theorem* 7.2.1.

8.3 Proof of theorem 7.3.1 (Draft)

Since $R_c(\hat{Y}^n) = nR$, we have:

$$\left|\hat{Y}^n(\mathbf{y}) - P^n(\mathbf{y})\right| \geq P^n(\mathbf{y}) \text{ if } P^n(\mathbf{y}) \leq \exp(-nR)/2$$

$$\sum_{\mathbf{y}\in\mathcal{Y}^n} \geq \sum_{\substack{\mathbf{y}\in\mathcal{Y}^n: \\ P^n(\mathbf{y})\leq\exp(-nR)/2}} P^n(\mathbf{y}) \tag{8.45}$$

$$\geq (n+1)^{-|\mathcal{Y}|} \exp\left\{-n \min_{\substack{V\in\mathcal{M}(\mathcal{Y}): \\ D(V||Y)+H(V)\geq R+1/n}}\right\} D(V||Y) \tag{8.46}$$

This implies that

$$d(P_Y^n, \hat{Y}_Y^n) = \sum_{\mathbf{y}\in\mathcal{Y}^n} \left|\hat{Y}(\mathbf{y}) - Y(\mathbf{y})\right| \tag{8.47}$$

$$\geq (n+1)^{-|\mathcal{Y}|} \exp\left\{-n \min_{\substack{V\in\mathcal{M}(\mathcal{Y}): \\ D(V||Y)+H(V)\geq R+1/n}} D(V||Y)\right\} \tag{8.48}$$

$$\geq (n+1)^{-|\mathcal{Y}|-1} \exp(-nE_{sp}(R+1/n, Y)) \tag{8.49}$$

Or

$$-\frac{1}{n}\log d(P_Y^n, \hat{Y}_Y^n) \geq (|\mathcal{Y}|+1)\frac{\log(n+1)}{n} + E_{sp}(R+1/n, Y) \tag{8.50}$$

Note that $E_{sp}(R, Y)$ is a continuous function of R have have the first part of Theorem 4 proven. Now we investigate when the equality holds. Rewrite $E_{\mathrm{r}}(R, Y)$ as

$$E_{\mathrm{r}}(R, Y) = \min\left[E_{sp}(R, Y), E_1(R, Y)\right] \tag{8.51}$$

where

$$E_1(R, Y) \stackrel{\text{def}}{=} \min_{\substack{V\in\mathcal{M}(\mathcal{Y}): \\ D(V||Y)+H(V)\leq R}} [\, R - H(V)] \tag{8.52}$$

Obviously

$$E_1(R, Y) \geq R - \log M \tag{8.53}$$

where we have the equality if and only if$H(V) = \log M$. That means V is uniform. We have:

$$D(V||Y) + H(V) = D(V_0||Y) + H(V_0) = R_0$$

Therefore, if $R \geq R_0$

$$E_{\mathrm{r}}(R, Y) = R - \log M$$

Because $-H(V), D(V||Y) + H(Y)$ are convex function, linear function respectively of V. Then for $R \leq R_0$, the minimum of $E_1(R, Y)$ can be attainted at its boundry, that is

$$E_1(R, Y) = \min_{\substack{V \in \mathcal{M}(\mathcal{Y}): \\ D(V||Y)+H(V)=R}} [R - H(V)] = \min_{\substack{V \in \mathcal{M}(\mathcal{Y}): \\ D(V||Y)+H(V)=R}} D(V)||Y)$$

From above equation, we have:

$$E_1(R, Y) \leq E_{sp}(R, Y) \tag{8.54}$$

whenever $R \leq R_0$.

8.4 Proof of theorem 7.4.1 (Draft)

Note that $\hat{P}^n(\mathbf{x}) \leq P^n(\mathbf{x})$ for $\mathbf{x} \neq 11 \cdots 1$, we have

$$2 - \sum_{\mathbf{x} \in \mathcal{X}^n} \left| \hat{P}^n(\mathbf{x}) - P^n(\mathbf{x}) \right| = \sum_{\mathbf{x} \in \mathcal{X}^n} (\hat{P}^n(\mathbf{x}) + P^n(\mathbf{x})) - \sum_{\mathbf{x} \in \mathcal{X}^n} \left| \hat{P}^n(\mathbf{x}) - P^n(\mathbf{x}) \right| \tag{8.55}$$

$$= \sum_{\mathbf{x} \in \mathcal{X}^n} \left(\hat{P}^n(\mathbf{x}) + P^n(\mathbf{x}) - \left| \hat{P}^n(\mathbf{x}) - P^n(\mathbf{x}) \right| \right) \tag{8.56}$$

$$= 2 \sum_{\mathbf{x} \in \mathcal{X}^n} \min \left(\hat{P}^n(\mathbf{x}), P^n(\mathbf{x}) \right) \tag{8.57}$$

$$\leq 2 \sum_{\substack{\mathbf{x} \in \mathcal{X}^n : \mathbf{x} \neq 11 \cdots 1: \\ P^n(\mathbf{x}) \geq \exp(-nR)}} P^n(\mathbf{x}) + 2P^n(11 \cdots 1) \tag{8.58}$$

$$\leq 2 \sum_{\substack{\mathbf{x} \in \mathcal{X}^n :: \\ P^n(\mathbf{x}) \geq \exp(-nR)}} P^n(\mathbf{x}) + 2 \exp(-nR) \tag{8.59}$$

The last inequality, when we include $\mathbf{x} = 11 \cdots 1$ into the $\sum$, it holds for both cases:

$$P^n(11 \cdots 1) \geq \exp(-nR)$$

and

$$P^n(11 \cdots 1) < \exp(-nR)$$

Therefore we have:

$$2 - d(P^n, \hat{P}^n) \leq 2 \sum_{\substack{\mathbf{x} \in \mathcal{X}^n: \\ P^n(\mathbf{x}) \geq \exp(-nR)}} P^n(\mathbf{x}) + \exp(-nR) \tag{8.60}$$

$$\leq \sum_{\substack{V \in \mathcal{P}_n(\mathcal{X}): \\ D(V||X|Q)+H(V|Q) \leq R}} \exp(-nD(V||X|Q)) \tag{8.61}$$

which implies:

$$\liminf_{n\to\infty}\left[-\frac{1}{n}\log\left(2-d(P^n,\hat{P}^n)\right)\right]\geq F(R,P) \tag{8.62}$$

On the other hand,

$$\begin{aligned} 2-\sum_{\mathbf{x}\in\mathcal{X}^n}|\hat{P}^n(\mathbf{x}-P^n(\mathbf{x})| &\geq 2\sum_{\substack{\mathbf{x}\in\mathcal{X}^n:\\ P^n(\mathbf{x})\geq 2\exp(-nR)}}\min(\hat{P}^n(\mathbf{x}),P^n(\mathbf{x})) && (8.63)\\ &\geq 2\sum_{\substack{\mathbf{x}\in\mathcal{X}^n:\\ P^n(\mathbf{x})\geq 2\exp(-nR)}}\frac{1}{3}P^n(\mathbf{x}) && (8.64)\end{aligned}$$

Then in a similar manner, we have:

$$\limsup_{n\to\infty}\left[-\frac{1}{n}\log\left(2-d(P^n,\hat{P}^n)\right)\right]\leq F(R,P) \tag{8.65}$$

Therefore

$$\lim\left[-\frac{1}{n}\log\left(2-d(P^n,\hat{P}^n)\right)\right]=F(R,P) \tag{8.66}$$

8.5 Proof of theorem 7.5.1 (Draft)

We will show the convere part first. We have:

$$\begin{aligned} 2-\sum_{\mathbf{y}\in\mathcal{Y}^n}|\overline{P}_Y^n(\mathbf{y})-P_Y^n(\mathbf{y})| &= 2\sum_{\mathbf{y}\in\mathcal{Y}^n}\min\left(\overline{Y}^n(\mathbf{y}),Y^n(\mathbf{y})\right) && (8.67)\\ &\leq 2\sum_{\substack{\mathbf{y}\in\mathcal{Y}^n:\\ \overline{P}_Y^n(\mathbf{y}>0}}P_Y^n(\mathbf{y}) && (8.68)\end{aligned}$$

Since $\overline{P}_Y^n$ has resolution nR, $\overline{P}_Y^n(\mathbf{y})$ must be positive for at most $\exp(nR)$ sequences in $\mathcal{Y}^n$. Hence we have:

$$\begin{aligned} 2-d(P_Y^n,\overline{P}_Y^n) &\leq 2\sum_{\substack{\mathbf{y}\in\mathcal{Y}^n:\\ \overline{P}_Y^n(\mathbf{y})>0}}P_Y^n(\mathbf{y}) && (8.69)\\ &\leq 2\sum_{V\in\mathcal{M}(\mathcal{Y})}\exp\left(-nD(V||Y)\right)\min\left\{1,\frac{\exp(nR)}{\exp(nH(V))}\right\} && (8.70)\\ &\leq 2(n+1)^{1+|\mathcal{Y}|}\exp\left(-n\min_{V\in\mathcal{M}(\mathcal{Y})}\left[D(V||Y)+|H(V)-R|^+\right]\right) && (8.71)\end{aligned}$$

The set

$$S = \{V \in \mathcal{M}(\mathcal{Y}) : H(V) \geq R\} \tag{8.72}$$

is convex. If we restrict the region V to S then $D(V||Y) + H(V) - R$ is a linear function of V. Then, we have:

$$\min_{V \in S} \left[D(V||Y) + |H(V) - R|^{+} \right] = \min_{\substack{V \in \mathcal{M}(\mathcal{Y}): \\ H(V) = R}} D(V||Y) \tag{8.73}$$

Thereofore, the region that takes the mimimum for V in (8.71) can be retricted to $H(V) \leq R$, and we obtain:

$$\liminf \left[-\frac{1}{n} \log \left\{ 2 - d(P_Y^n, \overline{P}_Y^n) \right\} \right] \geq G(R, Y) \tag{8.74}$$

On the other hand,

$$2 - \sum_{\mathbf{y} \in \mathcal{M}(\mathcal{Y})} |\overline{P}_Y^n(\mathbf{y} - P_Y^n(\mathbf{y})| \geq 2 \sum_{\substack{\mathbf{y} \in \mathcal{M}(\mathcal{Y}): \\ P_Y^n(\mathbf{y}) \geq 2 \exp(-nR)}} \min(\overline{P}_Y^n(\mathbf{y}), P_Y^n(\mathbf{y})) \tag{8.75}$$

$$\geq 2 \sum_{\substack{\mathbf{y} \in \mathcal{M}(\mathcal{Y}): \\ P_Y^n(\mathbf{y}) \geq 2 \exp(-nR)}} \frac{1}{3} P_Y^n(\mathbf{y}) \tag{8.76}$$

Then in a similar manner, we have:

$$\limsup \left[-\frac{1}{n} \log \left\{ 2 - d(P_Y^n, \overline{P}_Y^n \right\} \right] \leq G(R, Y) \tag{8.77}$$

Therefore from (8.74) and (8.77) we can conclude

$$\lim \left[-\frac{1}{n} \log \left\{ 2 - d(P_Y^n, \overline{P}_Y^n \right\} \right] = G(R, Y) \tag{8.78}$$

Chapter 9

Conclusion

謝辞

Chapter 10

ChangeLog

- 2006/01/16 Fix proof of theorem 2, second part.
- 2006/01/16 Rename indices $ij \to kj$ in Notations and Basic Defs.
- 2006/01/16 Fix appendix. Now it works. Fix subsection to section in chapter "fixed number of coin tosses".
- 2006/01/16 Type of Markov chain $\to$ Markov types of sequences. Statements of the problems. More on introduction(at the end). More Historical Review(at the end): more on Uyematsu + Kanaya, and, all "In this paper".
- 2006/01/16 Re divergence def. Def: stachastic matrix to transition matrix, $Q \to P$. On Markov type: $\mathbf{x} \to X^n$. chain to sequence. In IA: coin Z with $\mathbf{p} \to \mathbf{z}$, resolution to resolvability, $Q_h^{(m)} = \sum p_{ji}, \quad , q \to p$.
- 2006/01/16 Rewrite theorem 4,5,6(are they correct?).
- 2006/01/16 Proof of theorem 2.$\gamma_n \to \frac{1}{n} \log p_{X_1}/p_{X_n X_1}$. Put γ_n inside the exponential and get the form of $E_{\mathrm{r}}(R + \gamma_n, P)$.
- 2006/01/11 Add definition of *resolution problem*.
- 2006/01/10 First attempt in creating Introduction, Historical Review, State of problem.
- 2006/01/08 Change documentclass to book, change section to chapter.
- 2006/01/06 Make toc for presentation.
- 2005/12/28 Redefinite $V_n(\mathcal{X})$.
- 2005/12/27 Move "About Type" to section 2. Massive typo correction.

Chapter 11

TODO

- Proof Markov type related formulas.

Appendix A

Appendix

A.1 Markov type ralated formulas: Proof of LEMMA 4.0.14

A.2 Other tiny tiny thingys

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Index